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Hybrid Emulsion-Dispersion (HED) Polymerization Enables One-Pot Synthesis of Functional Janus Nanoparticles

Utsav Kumar Dey¹  | Ricardo Ortega¹ | Serkan Demirci¹  | Yifan Li¹ | Fei Liu¹ | Zhengtao Yang¹ | Tiago Twedt¹ | Prabhdeesh Singh² | Juan-Carlos Mora-Diaz³ | Nubia Resende De Macedo³ | Luis G. Gimenez-Lirola³ | Shan Jiang¹ 

¹Department of Materials Science and Engineering, Iowa State University of Science and Technology, Ames, Iowa, USA | ²Department of Biochemistry, Biophysics and Molecular Biology, Iowa State University of Science and Technology, Ames, Iowa, USA | ³Veterinary Diagnostic and Production Animal Medicine, Iowa State University of Science and Technology, Ames, Iowa, USA

Correspondence: Shan Jiang (sjiang1@iastate.edu)

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ABSTRACT

Janus nanoparticles possess anisotropic architectures that enable strong interfacial interactions and enhanced functionality, yet their synthesis often depends on multistep seeded or templated routes. Here we report hybrid emulsion-dispersion (HED) polymerization, a one-pot strategy that introduces a fundamentally different route to Janus morphology by inducing symmetry breaking via polymerization-induced phase separation in a single reactor. This process directly yields polymeric Janus nanoparticles without seed preparation, sequential monomer addition, or post-modification. Kinetic studies and a morphology map reveal that symmetry breaking is established within the first minutes of polymerization, while the final morphology is governed by the balance between solubility, interfacial energy, and kinetic arrest. This strategy produces sub-200 nm Janus nanoparticles with tunable morphologies and compositions. As a proof of concept, we synthesize N-halamine-functional Janus nanoparticles via HED and demonstrate their effectiveness in coatings. The particles preferentially enrich at coating interfaces and enable efficient, rechargeable antibacterial activity.

1 | Introduction

Janus particles are anisotropic colloids that integrate two distinct domains of different chemistry within a single particle. The Janus geometry gives rise to preferential adsorption at interfaces, directional assemblies, and multiple functionalities [1–3]. Many of the unique properties are inaccessible to conventional isotropic particles, making Janus particles attractive for applications such as emulsion stabilization [4, 5], catalysis [6, 7], coatings [8, 9], and drug delivery [10]. Over the past two decades, extensive studies have been dedicated to synthesis [11–13], self-assembly [14, 15], and applications [16, 17] of Janus particles. Despite various demonstrations, translation into scalable and industrial applications remains very limited. A primary bottleneck is

synthetic complexity. Most polymeric Janus nanoparticles are produced through multistep seeded routes that require preparation of seed particles, controlled swelling with a second monomer, and subsequent polymerization [18, 19]. These strategies demand repeated purification, tight control over intermediate composition, and narrow processing windows. Minor variations in seed quality [20, 21], concentration [22], or swelling conditions [23–26] often result in loss of anisotropy. The challenges are amplified with smaller particles at the submicron scale, where colloidal stability, rapid kinetics, and seed separation become increasingly difficult to manage [27]. As a consequence, robust one-pot approaches that directly generate functional polymeric Janus nanoparticles remain rare [11, 28, 29].

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Equally important is the identification of applications in which Janus geometry provides a clear advantage. Recent studies show that when added as a coating additive, Janus particles can preferentially stratify toward coating surfaces during the film drying process, generating layered structures that enrich functional groups at the air interface [9, 30]. This interfacial localization is particularly attractive for antibacterial coatings, where activity is required at the surface rather than in the bulk of the film. By concentrating antibacterial functionality at the region of action, Janus particles can enhance surface efficacy while reducing the total additive loading. In this manner, surface performance can be enhanced while simultaneously reducing material consumption and cost.

Contaminated surfaces contribute significantly to the transmission of infectious diseases in hospitals and long-term care facilities, with environmental pathways estimated to account for approximately one quarter of cases [31–33]. Durable self-sanitizing coatings are therefore of substantial societal and economic interest. Conventional antibacterial technologies, including leaching biocides, metal-based additives, and contact killing agents, often suffer from limited lifetime, toxicity concerns, or the risk of antibacterial resistance development [34–37]. N-halamine systems have emerged as an attractive alternative due to their broad-spectrum efficacy, low toxicity, minimal resistance, and chemical rechargeability [38–41]. However, their limited water solubility and high cost have hindered incorporation into waterborne coatings and limited large-scale deployment [42].

Here, we introduce hybrid emulsion-dispersion (HED) polymerization as a scalable, single-reactor, one-pot route to synthesize functional antibacterial Janus nanoparticles. HED polymerization integrates features of emulsion and dispersion polymerization within one pot, enabling concurrent polymerization of monomers with different solubilities. In HED polymerization, monomers, initiator, stabilizer, and solvent are present before polymerization begins. No preformed seed is introduced, and no discrete seed-isolation, swelling, sequential monomer-addition, or post-modification step is used. Anisotropy arises from polymerization-induced symmetry breaking during the earliest stage of particle growth, rather than from templated seeds or sequential monomer addition. This distinction eliminates the need for preformed seeds, swelling steps, or post-synthetic modification. Using this HED framework, we synthesized Janus nanoparticles composed of polystyrene and a hydantoin-based N-halamine monomer. The resulting particles exhibit well-defined dumbbell and snowman morphologies and preserve antibacterial functionality. When incorporated into waterborne coatings, these Janus particles preferentially stratify to the coating air interface, substantially enriching surface accessible N-halamine groups. This spatial enrichment enables efficient and rechargeable antibacterial performance at low additive loadings.

This work establishes HED polymerization as a fundamentally distinct concept for programming anisotropy and functionality into polymeric nanoparticles. Future work will extend the methodology to other functional Janus particles beyond the chemistry demonstrated here. By coupling scalable synthesis with application-driven interfacial design, it advances Janus particles from model colloids toward more practical material platforms.

2 | Results and Discussion

2.1 | Hybrid Emulsion-Dispersion (HED) Polymerization

Conventional strategies for synthesizing polymeric Janus particles by emulsion polymerization typically rely on multistep protocols that involve seed preparation, second monomer swelling, and subsequent polymerization. The final morphology is highly sensitive to intermediate conditions. Oftentimes, small deviations in seed concentration or swelling conditions can disrupt the formation of Janus morphology and lead to homogeneous particles or ill-defined particles [23, 43]. These challenges greatly limit scalability and reproducibility. In contrast, the approach presented here loads both monomers into a single reactor and enables direct formation of Janus architecture without preformed seeds or sequential processing (Figure 1a). In a typical experiment, hydantoin acrylamide (HA) and styrene were copolymerized at 80°C using potassium persulfate (KPS) as an initiator and polyvinylpyrrolidone (PVP) as a stabilizer. Under optimized conditions, highly monodisperse Janus particles of poly(styrene-co-hydantoin acrylamide) or P(S-co-HA) were obtained with either dumbbell or snowman morphologies (Figure 1b and Figure S1).

Fourier transform infrared (FTIR) spectra confirm the presence of characteristic vibrational bands from both PS and PHA in the Janus particles (Figure S2), consistent with incorporation of both polymers. High-resolution field emission scanning electron microscopy (FESEM) and scanning transmission electron microscopy (STEM) images further reveal a distinct contrast in surface texture between the two lobes (Figures S3 and S4). One domain exhibits a smooth surface, whereas the opposing domain displays a markedly rough morphology. Independent homopolymerization of HA produces PHA particles with rough surfaces, while homopolymerization of styrene yields PS particles with smooth surfaces. This clear morphological distinction strongly indicates that the smooth lobe of the Janus particles corresponds to PS, while the rough lobe is composed of PHA. The anisotropic chemical distribution of the Janus particles was further confirmed by TEM-EDX elemental mapping of chlorinated particles, which showed preferential localization of chlorine in one lobe of the dumbbell morphology (Figures S5 and S6). Selective etching of the PHA-rich domain with trifluoroacetic acid completely removes one lobe and leaves behind a single spherical particle, thereby confirming the compositional separation between the two domains (Figure S7).

Because HA exhibits limited solubility in pure water, polymerization was conducted in an ethanol (EtOH)-water medium to improve HA solubility. Under these conditions, styrene remains largely insoluble and exists as a separate hydrophobic phase that acts as a monomer reservoir for particle growth (Figure S8). In this solvent system, HA follows a dispersion polymerization pathway in the continuous phase, whereas styrene proceeds through an emulsion-type mechanism. When both monomers are introduced simultaneously, these two mechanistically distinct polymerizations proceed in parallel within a single reactor and are intrinsically coupled through interfacial adsorption,

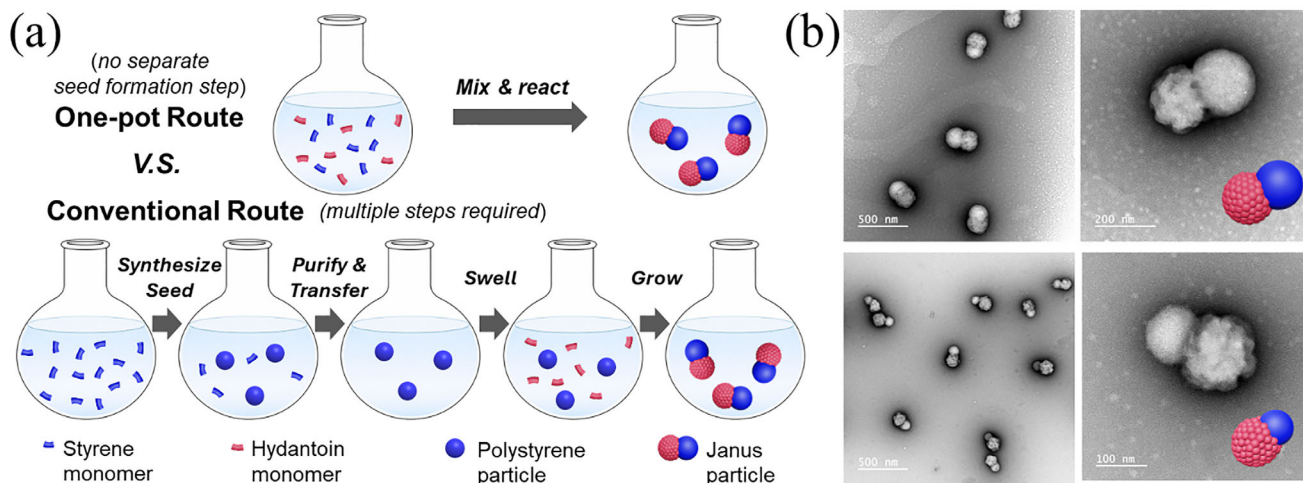


FIGURE 1 | (a) Schematic comparison between one-step route (this work) and the conventional multistep polymerization route, including seed synthesis, second monomer mixing followed by swelling, and polymerization at the end; (b) STEM images of the snowman (top) and dumbbell (bottom) Janus particles synthesized via the one-step route.

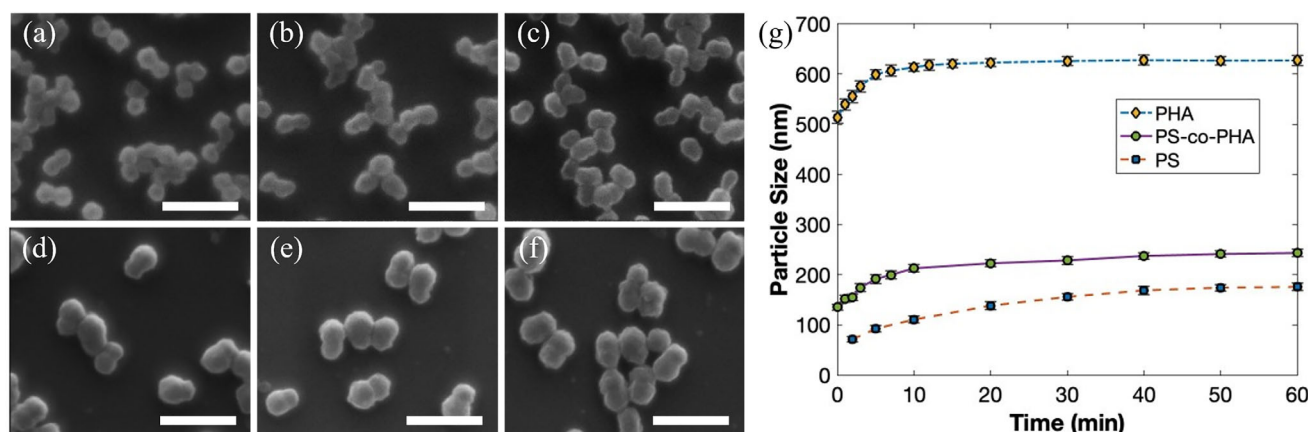


FIGURE 2 | P(S-co-HA) Janus particle growth after reaction reaching 80°C for (a) 1 min, (b) 3 min, (c) 5 min, (d) 10 min, (e) 20 min, and (f) 40 min (Scale bar: 500 nm); (g) Comparing controlled particle growth of P(S-co-HA) Janus particles with PS and PHA in 4/1 water/EtOH mixture, analyzed via DLS.

swelling, and co-growth. We name this one-pot hybrid emulsion dispersion polymerization or HED polymerization. The fundamental distinction between HED and conventional seeded emulsion approaches lies in the requirement of separated pre-established seeds, swelling steps, or sequential monomer dosing. In HED polymerization, all reactants were mixed together without any need for separation during the reaction. The interaction of emulsion nucleation and dispersion precipitation during the earliest stage of polymerization drives spontaneous symmetry breaking and directly yields Janus morphologies.

2.2 | HED Polymerization Kinetics and Mechanisms

To gain insight into the reaction mechanism, aliquots were withdrawn between 1 min and 4 h after reaching 80°C to study particle morphology (Figure 2a–f). Within 1 min, irregular nanoparticles

smaller than 150 nm are observed. By 3 min, two domain structures start to emerge, and by 5 min, well-defined Janus morphologies are observed. Subsequent growth proceeds primarily through enlargement of the two established lobes rather than nucleation of additional domains. These observations indicate that phase separation is initiated within the first few minutes of polymerization. Dynamic light scattering (DLS) further clarifies the growth behavior. Compared with homopolymerizations, HED produces particles whose sizes fall between those of pure PS emulsion polymerization and pure HA dispersion polymerization (Figure 2g). Styrene homopolymerization yields small particles due to initiation within small micelle-like clusters. In contrast, HA homopolymerization leads to larger particles, ~500 nm, consistent with precipitation only after oligomers reach a critical size in dispersion polymerization. The intermediate particle size observed in HED results from the coupled mechanisms of conventional emulsion polymerization and dispersion polymerization. In this process, PS first grows from small micelle-like structures

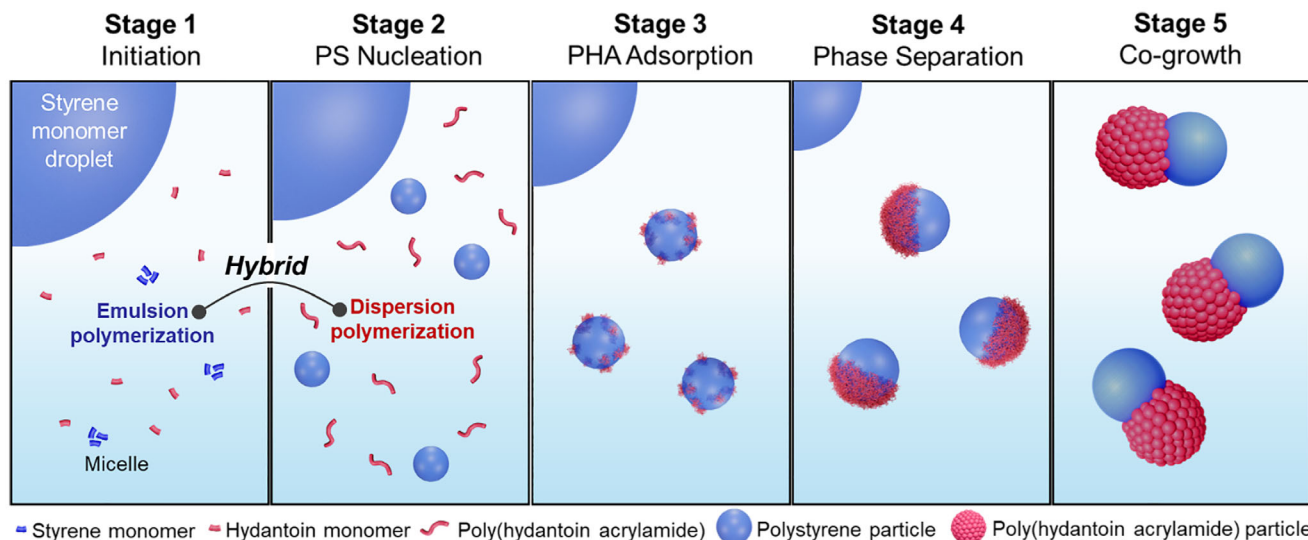


FIGURE 3 | Schematic illustration of the proposed formation mechanism of the P(S-co-HA) Janus particles: Step 1. Initial stage; Step 2. Primary nucleation of PS-rich oligomeric particles upon limited solubility of growing chains; Step 3. Diffusion of HA monomer toward and around the styrene nuclei; Step 4. Onset of polymerization-induced phase separation at the styrene-HA interface; Step 5. Concurrent polymerization and co-growth of the two lobes at the evolving interface until the completion of polymerization in both domains.

and serves as the initial seed. Subsequently, PHA adsorbs onto the PS seeds, followed by further polymerization on their surfaces.

Nuclear Magnetic Resonance (NMR) spectroscopy provides quantitative insight into monomer conversion. At 80°C, styrene is consumed more rapidly during the initial stage, indicating preferential formation of PS-rich nuclei (Figures S9 and S10). In contrast, within 1 h, HA reaches 98% conversion, whereas styrene reaches 52% (Figure S11a). This inversion of relative conversion rates can be rationalized by local monomer concentration effects. Early in the reaction, styrene polymerizes within micelle-like structures where its effective concentration is high. On the other hand, although HA is intrinsically more reactive, it is distributed throughout the continuous phase and much more dilute. As HA oligomers form and associate with PS nuclei, their local concentration increases and PHA growth accelerates. Experiments conducted at 50°C further show that styrene initiates and propagates more readily at lower temperature than HA, reinforcing the conclusion that PS seeds form first and act as nucleation sites for subsequent HA deposition (Figure S11b). Control experiments further underscore the necessity of simultaneous hybrid polymerization conditions. When HA is injected into a styrene seed solution, only spherical particles were obtained (Figure S12). When styrene and HA were polymerized separately and then mixed shortly after initiation, highly heterogeneous aggregates formed (Figure S13). These results demonstrate that carrying out the two polymerization reactions in the same pot from the very beginning is not only convenient, but also essential for reproducible Janus formation.

Based on these observations, we propose a five-stage mechanism (Figure 3). First, styrene nucleates within micelle-like clusters and forms PS seeds within ~1 min. Second, HA undergoes dispersion polymerization to form soluble oligomers. Third, HA monomers and oligomers adsorb and swell the PS seeds. Fourth,

polymerization-induced phase separation occurs at the evolving PS/PHA interface within approximately 5 min. Fifth, continued polymerization in each domain leads to concurrent lobe growth until monomer depletion. Because adsorption and phase separation occur rapidly, the seed stage is very difficult to isolate experimentally, highlighting the intrinsically coupled nature of HED polymerization.

2.3 | HED Polymerization Morphology Map

To establish structure-process relationships, we constructed a morphology map as a function of ethanol fraction and PVP concentration while keeping other parameters constant (Figure 4 and Figures S14–S17). Particles were classified by SEM into the following categories: sphere, acorn, multilobe, dumbbell, snowman, and mixed rough morphology. A workflow for quantitatively analyzing morphology is illustrated in Supporting Information (Figures S18 and S19). This diagram does not represent thermodynamic equilibrium. Rather, it captures the morphology of nanoparticles produced under coupled thermodynamic and kinetic conditions.

The role of ethanol is examined first. Ethanol modifies both monomer solubility and interfacial energetics. Solubility measurements confirm that HA solubility increases with ethanol content. In HA homopolymerization, increased solubility delays precipitation and enhances chain mobility. In HED polymerization, this directly affects PHA domain formation and its ability to coarsen. Once PS and PHA segregate, the accessible morphologies are constrained by minimization of total interfacial free energy: $G = \gamma_{12}A_{12} + \gamma_{23}A_{23} + \gamma_{13}A_{13}$, where γ_{ij} and A_{ij} represent interfacial tension and interfacial area between PS (1), medium (2), and PHA (3) phases [44]. However, achieving a low-energy configuration requires sufficient mobility before kinetic arrest.

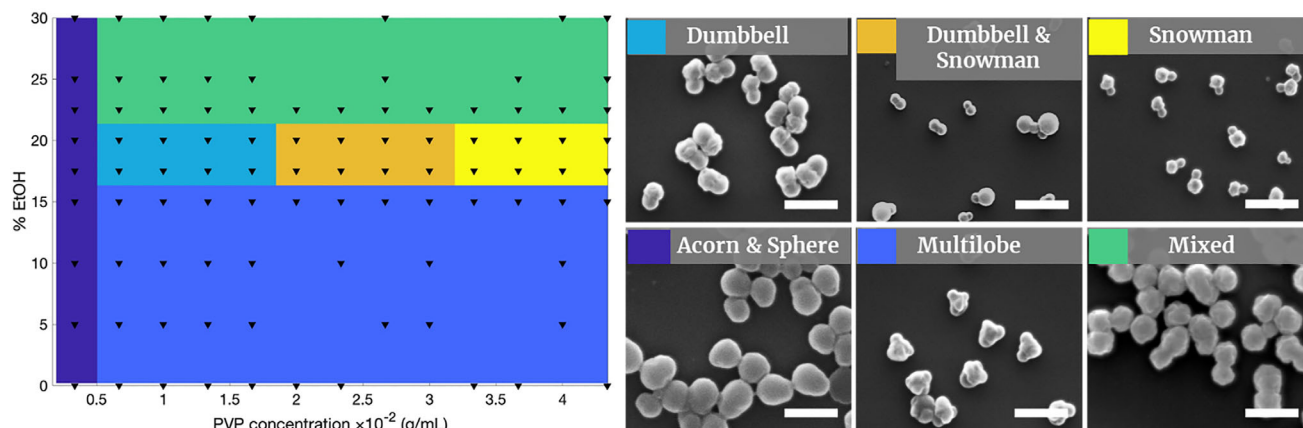


FIGURE 4 | Morphological phase map illustrating the dependence of Janus particle formation on key synthesis parameters, including stabilizer (PVP) concentration and solvent composition (%EtOH). Colored regions denote distinct particle morphologies, with representative FESEM images shown on the right (scale bar: 500 nm).

At low ethanol fractions $< 15\%$, limited HA solubility leads to rapid precipitation and insufficient relaxation time. Multilobe structures dominate, consistent with kinetically trapped intermediates (Figure S20a–c). At intermediate ethanol fractions $\sim 20\%$, delayed precipitation allows coalescence into a single PHA domain per PS seed, yielding well-defined Janus particles. At high ethanol fractions $\sim 25\%$, rough mixed morphologies emerge (Figure 4 and Figure S20e). Here, increased solvation, reduced γ_{12} and γ_{23} together with altered balances may reduce the free energy incentive to a clear phase separation. The transition is therefore governed by a shift from kinetic limitation to interfacial energy balance.

The role of PVP is also critical [45]. Within the ethanol window that produces Janus particles, PVP primarily controls nucleation density and interfacial stabilization. Increasing PVP reduces PS seed size, consistent with stabilizer-controlled nucleation. The smaller seed particles make it easier to shift morphology from symmetric dumbbell to asymmetric snowman as the PHA lobe grows (Figure S21). Importantly, PHA remains a single lobe across this range, indicating that PVP does not alter the fundamental phase separation pathway. The dumbbell to snowman transition can be interpreted as redistribution of interfacial areas between two domains whose interfacial penalties toward the medium are comparable. At intermediate PVP levels, both morphologies coexist, suggesting operation near a shallow free energy minimum where small variations in surface coverage are sufficient to bias the final configuration. At sufficiently high PVP concentrations, gelation of the medium is observed experimentally, indicating that increased viscosity and network formation may also arrest morphological relaxation. Throughout the morphology map, the styrene/HA weight ratio was maintained constant to isolate the effects of solvent composition and PVP concentration. Additional experiments performed under a representative synthesis condition (0.02 g/mL PVP and 20 vol% ethanol) demonstrate that varying the styrene/HA molar ratio also influences particle morphology (Figure S22), indicating that monomer composition represents another important parameter for controlling Janus particle formation.

To evaluate the scalability of the HED polymerization, the synthesis was performed at progressively larger batch volumes of 15, 100, and 250 mL while maintaining the same formulation and reaction conditions. Comparable Janus particle morphologies were obtained at all three reaction volumes (Figure S23), demonstrating that the process can be scalable.

Collectively, the analysis of the morphology map, together with thermodynamics and kinetics considerations, demonstrates the different roles of ethanol and PVP. The ethanol amount defines the kinetic window for PHA domain formation by regulating solubility and mobility, whereas PVP tunes PS nucleation and alters interfacial area allocation. Morphology selection in HED polymerization therefore arises from a coupled thermodynamic-kinetic competition, enabling robust and scalable synthesis of sub-200 nm Janus particles without multistep processing.

2.4 | Stratified Coatings and Antibacterial Performance

Having established the P(S-co-HA) Janus morphology, we next evaluate these particles as coating additives. Previous studies have shown that highly amphiphilic Janus particles can preferentially stratify toward coating surfaces during film formation [9]. Here, we demonstrate that Janus particles bearing N-halamine functionality exhibit similar stratification behavior (Figure 5 and Figure S24), despite their comparatively lower amphiphilicity. This surface enrichment enables the Janus particles to carry antibacterial functional groups to the coating interface, thereby maximizing interfacial exposure and enhancing antibacterial efficacy. Such stratification also allows effective antibacterial performance at relatively low additive loadings, reducing material consumption and overall formulation cost without compromising functionality. In formulations employing $\sim 1.4 \mu\text{m}$ homogeneous spherical particles as binders, coatings containing 2 to 10 wt.% Janus particles exhibited nearly complete surface coverage by Janus particles after drying (Figure 5c–f and Figure S25).

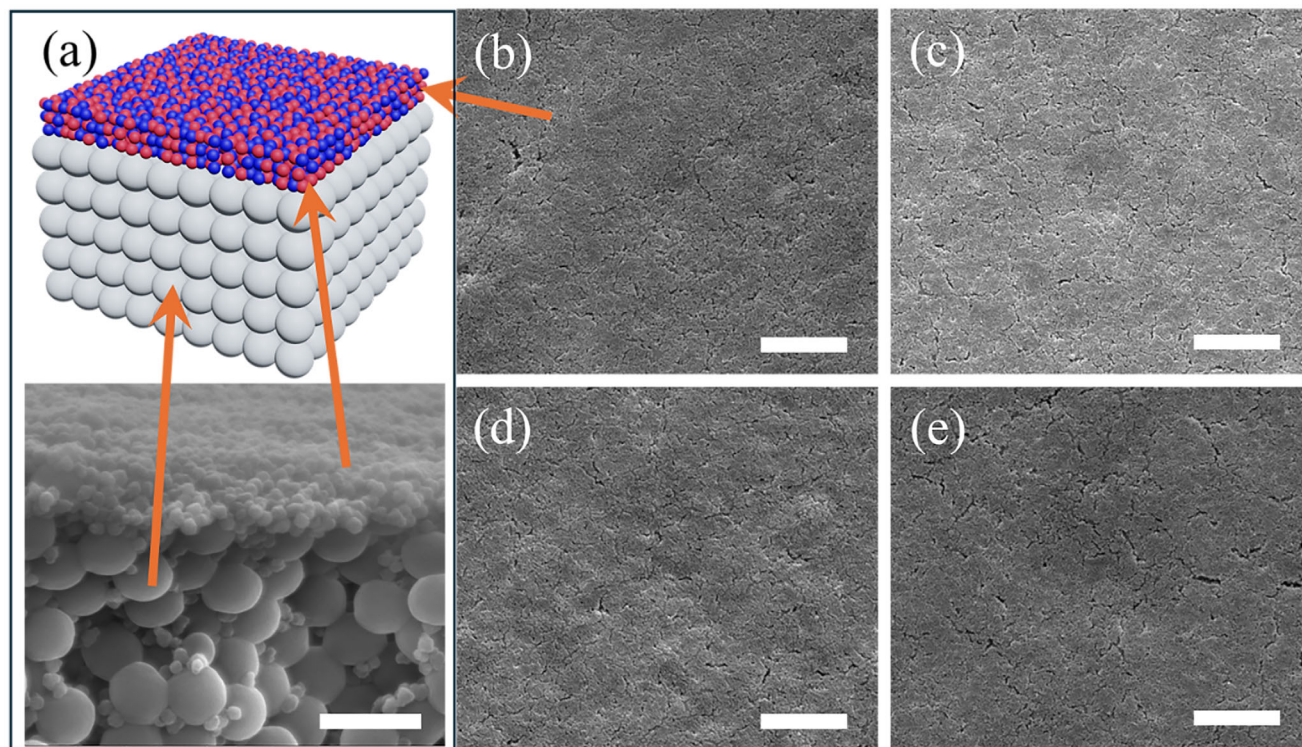


FIGURE 5 | (a) Schematic illustration of a self-stratified coating with cross-section view of the coating under SEM, where Janus particles migrate to the surface while the polymer binder forms the underlying layer (scale bar: 2 μm); Top-view FESEM images of coatings incorporating (c) 2%, (d) 5%, (e) 7%, and (f) 10% Janus particles as additives (scale bar: 5 μm).

We further found that stratification is sensitive to coating pH. When the formulation pH was adjusted to basic (pH 11), full coverage of the coating surface with Janus particles was observed. In contrast, formulations with pH adjusted to acidic (pH 4) had their Janus stratification reduced (Figure S26). The reduction in stratification is mainly achieved due to changes in surface charge and dispersion stability. At basic pH, hydantoin N–H groups can partially deprotonate, giving the particle surface higher negative charge. This increases electrostatic repulsion, keeps particles well-dispersed during drying, and allows them to migrate and stratify effectively at the surface. As pH shifts from neutral to acidic, the hydantoin groups remain protonated (neutral), resulting in lower surface charges, weakened repulsion, and increased particle aggregation, reducing mobility and leading to much weaker surface stratification. This is consistent with previous studies investigating the role of formulation pH and surface charge on the amount of Janus particles that successfully stratify to the coating surface after film drying [9, 46].

Quantification of oxidative chlorine content shows an approximately linear increase from 0.39×10^{17} atoms/cm² at 2 wt.% to 2.76×10^{17} atoms/cm² at 10 wt.% (Figure 6a), suggesting multilayer formation near surfaces. Rechargeability studies reveal only a modest decline in performance over successive discharge and recharge cycles. After five cycles, the coatings retain 65% to 79% of their initial chlorine content, underscoring the robustness of the system and its ability to withstand repeated chlorination (Figure 6b). Such behavior suggests that coatings incorporating Janus particles can tolerate periodic bleach treatment and sanitation procedures without substantial loss of functionality, a critical requirement for practical deployment.

These films were subsequently evaluated for antibacterial performance. Coating films containing pure binder particles (0 wt.% Janus particles), 3 wt.% Janus particles, and 5 wt.% Janus particles were first chlorinated and then exposed to bacterial suspensions of *Staphylococcus Aureus* (*S. aureus*) and *Escherichia coli* (*E. coli*) for a contact time of 30 min. After exposure, residual active chlorine was quenched with sodium thiosulfate, and aliquots were drop cast onto nutrient agar plates to quantify viable colony-forming units (CFUs). The pure binder coating exhibited bacterial counts of approximately 5.08 and 5.23 log₁₀ CFU mL⁻¹ for *S. aureus* and *E. coli*, respectively, confirming negligible intrinsic antibacterial activity. In contrast, fully stratified coatings incorporating 3 wt.% Janus particles showed a substantial reduction in viable bacteria, ranging from 66% to 100% reduction depending on the strain. Partially stratified coatings at the same loading displayed lower antibacterial efficacy, consistent with reduced surface availability of N-halamine groups. At 5 wt.% Janus particle loading, antibacterial performance was further enhanced. Fully stratified coatings achieved complete inactivation, corresponding to 100% reduction of both bacterial strains (Figure 6c,d), whereas partially stratified coatings again exhibited lesser performance. These results directly correlate antibacterial efficacy with the degree of surface stratification, highlighting the importance of interfacial enrichment in maximizing functional performance.

To further support the ability of stratified coatings to sanitize bacterial strains on their surface, Student's t-test statistical analysis was done in comparison to the control case containing 0 wt.% Janus particles to determine whether reductions in viable bacteria were statistically significant. For antibacterial experiments against *S. aureus*, both 3 and 5 wt.% stratified coatings were

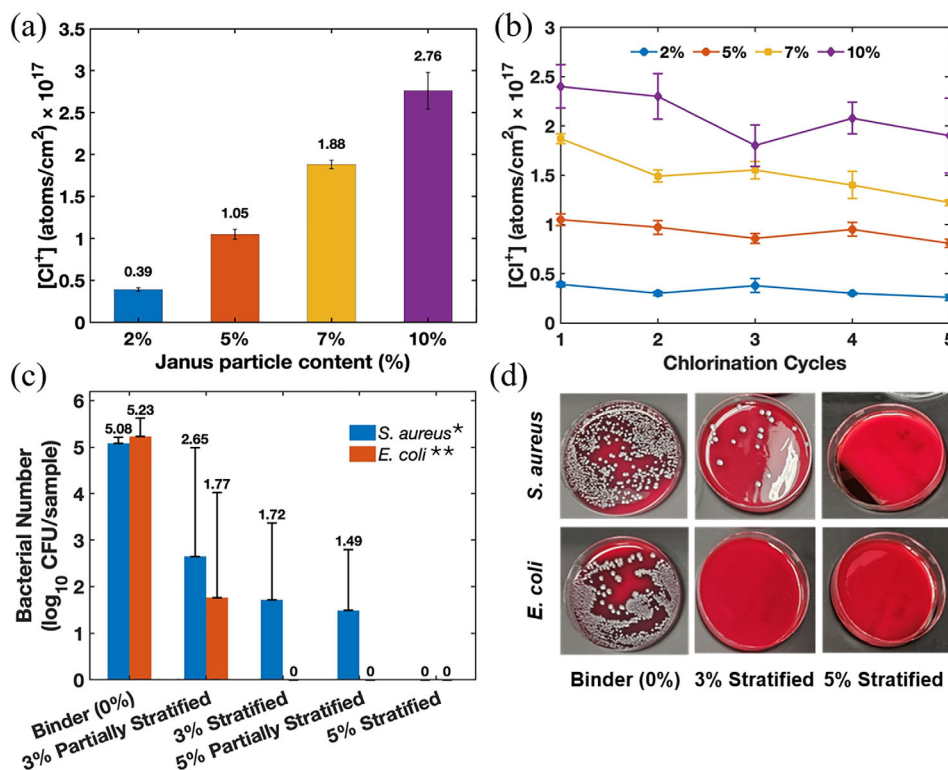


FIGURE 6 | (a) Comparison of oxidative chlorine of coatings with varying Janus particle concentrations, (b) Oxidative chlorine rechargeability of stratified coatings with 2, 5, 7 and 10% Janus particle additive, (c) Antibacterial testing of the coatings with 0% (binder control), 3% (partially stratified and stratified) and 5% (partially stratified and stratified) Janus particle additive against gram-positive *S. aureus* and gram-negative *E. coli*. 3% & 5% stratified samples compared against the control case (0% JP) showed statistical significance for both *S. aureus* ($p < 0.05$ and $p < 0.001$, respectively) and *E. coli* ($p < 0.01$ and $p < 0.01$, respectively) experiments. (d) Resulting agar plates showing bacterial colony formation.

found to significantly reduce bacteria on the surface within a 95% confidence level ($p < 0.05$ and $p < 0.001$, respectively). In experiments involving *E. coli*, the reduction was found to be even greater for stratified coatings at both additive loadings ($p < 0.01$ for both 3 and 5 wt.% Janus particles). This analysis supports the stratified coating's ability to effectively reduce both bacterial strains upon direct contact with active N-halamines on their surface.

This marked improvement is consistent with prior reports on N-halamine systems. Antibacterial activity originates from the N–Cl functionality, where oxidative “Cl⁺” species induce oxidative damage to bacterial cell walls in a manner analogous to hypochlorite (ClO[−]). During this process, Cl⁺ is reduced and released as Cl[−], resulting in depletion of active oxidative sites. Importantly, such chlorine loss underlies the rechargeability of N-halamine coatings, as subsequent chlorination restores the N–Cl bonds and regenerates antibacterial efficacy. This contact-killing behavior has been correlated with surface-accessible oxidative chlorine contents typically in the order of $\sim 10^{15}$ atoms cm^{−2} [47]. In our fully stratified coatings containing 5 wt.% Janus particles, chlorine densities reached $\sim 10^{17}$ atoms cm^{−2}, consistent with the observed complete bacterial inactivation.

It is important to emphasize that this study is intended to demonstrate proof of concept for Janus-based antibacterial coatings. Current formulations were deliberately simplified and

do not include systematic optimization of film formation or adhesion. These factors likely contribute to the relatively large error bars observed in the antibacterial measurements. Further refinement of binder composition, film formation, and interfacial adhesion would be expected to improve coating uniformity and enhance the consistency and reproducibility of antibacterial performance.

3 | Conclusion

This work establishes HED polymerization as a straightforward yet powerful strategy for generating polymeric Janus nanoparticles through a one-pot reaction. By coupling emulsion nucleation of hydrophobic monomers (styrene) with dispersion polymerization of more soluble monomers (HA) in the same reactor, chemical anisotropy emerges spontaneously during the earliest stages of particle growth. Unlike conventional seeded or multistep approaches, this method eliminates seed isolation, swelling steps, and sequential processing, offering a streamlined and scalable pathway to sub-200 nm Janus nanoparticles with controllable morphology. More importantly, we demonstrate that the functional antibacterial Janus nanoparticles can be produced using HED polymerization. When incorporated into waterborne coatings, the Janus nanoparticles spontaneously migrate toward the air interface during film drying, concentrating N-halamine groups where antibacterial action is required. This stratification enriches surface accessible oxidative chlorine to $\sim 10^{17}$ atoms

cm⁻², enabling complete inactivation of *S. aureus* and *E. coli* at only 5 wt.% loading. The coatings further retain a significant amount of oxidative chlorine capacity after five recharge cycles, demonstrating potential for renewable antibacterial activity.

Although the HED polymerization strategy successfully produced functional Janus nanoparticles in this study, several limitations should be noted. The present work investigates only the styrene/HA monomer system. In addition, particle morphology was found to be sensitive to solvent composition, stabilizer concentration, and monomer ratio. Finally, the coating stratification and antibacterial performance can be further improved by optimizing binders and coating formulations. These limitations also represent important directions for future investigation.

Beyond the specific P(S-co-HA) system demonstrated here, HED polymerization provides a versatile and adaptable framework for constructing anisotropic polymer nanoparticles from monomers with contrasting solubility, reactivity, and interfacial affinity. The concept of coupling distinct polymerization pathways within a single reaction environment can be extended to other functional monomers, including systems incorporating stimuli-responsive, catalytic, electronic, or bioactive chemistries. More broadly, this work demonstrates that symmetry breaking during early-stage polymerization can be intentionally harnessed as a synthetic design principle. By leveraging this concept, HED polymerization expands the accessible design space of polymer nanoparticles beyond what is achievable through conventional seeded or sequential methods. This framework opens new opportunities not only for the scalable production of multifunctional Janus nanomaterials, but also for translating nanoscale anisotropy into advanced coatings, responsive interfaces, and novel soft matter systems.

Author Contributions

U.K.D. performed particle synthesis and characterization, coating formulation, chlorine analysis, rechargeability studies, antibacterial testing, and mechanistic analysis. R.O. performed coating formulation and characterization, chlorine analysis, rechargeability studies, and antibacterial testing. S.D. conducted NMR analysis, monomer conversion measurements, and mechanistic studies, and contributed to particle synthesis and characterization. F.L. contributed to figure preparation and manuscript revision. Z.Y. contributed to particle synthesis and antibacterial testing. T.T. contributed to synthesis and antibacterial tests. P.S. contributed to synthesis. J.C.M.D., N.R.D.M., and L.G.G.L. performed antibacterial experiments. S.J. conceived the concept, supervised the project, and wrote and revised the manuscript. All authors reviewed and approved the final version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon request.

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Supporting Information

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